

Membrane Transport in Brief

LEFT COLUMN START. Every living cell is separated from its surroundings by a thin boundary that controls what enters and what leaves. This boundary is not a rigid wall but a flexible, self-sealing film that constantly rearranges itself as the cell grows, divides and responds to signals. Researchers first described the boundary as a simple film of oil, but careful measurements in the twentieth century showed that it has a defined thickness and a layered structure. Those measurements led to the models that are still taught in introductory courses today.

Proteins embedded in the boundary act as channels, pumps and receptors. Some span the full width of the film, while others sit on only one face. Their arrangement explains why the two faces of the boundary behave differently. LEFT COLUMN END.

RIGHT COLUMN START. Small, uncharged molecules such as oxygen cross the boundary easily, whereas ions and large molecules need help. The cell spends a large share of its energy budget on moving these substances against their natural direction of flow. When the balance of water on either side changes, the cell may swell or shrink. Plants, fungi and many bacteria resist bursting because a stiff outer wall surrounds the softer inner boundary.

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